KYLE KAMAY

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EDUCATION

University of Connecticut, Storrs, CT
Honors Bachelor of Science, May 2029
Major: Electrical Engineering; GPA: 3.68

TECHNICAL SKILLS

Programming: Python, C++,
Hardware: Arduino, ESP32, Sensor Integration, Circuit Analysis, Breadboarding
Software: GitHub, Microsoft Office

Engineering Experience

UConn Tech Park – Travelers Project Challenge, Hartford, Connecticut
Engineering Project Participant, May 2026 - September 2026

- Developed a Python-based physics and statistical model to estimate hail damage probability across multiple material types using impact energy, material properties, and inspection data
- Implemented Monte Carlo simulations to model randomized storm conditions and compare predicted damage risk across roofing, automotive, and structural materials
- Conducted technical and competitive research on weather patterns, hail dynamics, drone technology, and industry approaches to improve property-damage assessment
- Collaborated with Travelers professionals to analyze existing damage-assessment processes and developed a web-based platform to communicate research findings, technical analysis, and recommendations

UConn Nursing & Engineering Innovation Center, Storrs, Connecticut
NurseEng Fellow, August 2025 - May 2026

- Designed and prototyped a pressure-sensing mattress pad to monitor patient pressure distribution and support pressure-injury prevention
- Integrated force-sensitive resistor sensors with embedded microcontroller hardware to collect and analyze real-time pressure data
- Collaborated with nursing students and faculty to turn clinical needs into technical requirements and iterate on a functional prototype

Engineering Projects

Embedded Patient Pressure Monitoring System

- Integrated FSR sensors, FireBeetle ESP32, analog multiplexing, and Bluetooth communication to acquire and transmit data
- Iterated on sensor placement, hardware architecture, and data acquisition methods to improve system scalability
- Developing a real-time pressure heat map to visualize sensor output and identify areas of elevated pressure

Physics-Based Hail Damage Risk Modeling & Simulation

- Developed a Python physics model calculating hail mass, velocity, and impact energy to estimate material damage risk
- Implemented Monte Carlo simulations to generate randomized storm conditions and estimate damage probabilities across materials
- Developed data visualizations and material risk rankings to analyze relationships between hail size, impact energy, and predicted probability

Low-Cost Heart Rate Monitoring System

- Designed and prototyped a wearable-style heart rate monitoring system using a pulse sensor and Arduino Uno for real-time physiological data acquisition
- Programmed Arduino/C++ software to collect, process, visualize, and interpret heart rate measurements
- Tested sensor placement and acquisition methods to improve signal consistency and measurement reliability

Leadership & Activities

IEEE – Member **| UConn Underwater Robotics –** Member **| UConn Formula One – EV** – Contributor